

Q7(a)

Simple harmonic motion refers to the oscillatory motion of an object about an equilibrium position such that its acceleration is proportional to its displacement and is always directed towards the equilibrium position.

7(b)(i)

Angular frequency ω is 2π times the frequency of an object's oscillatory motion, i.e. $\omega = 2\pi f$.

It is related to the period T by the equation $\omega = \frac{2\pi}{T}$

7(b)(ii)

Given $\omega^2 = \frac{g}{l}$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{l}{g}}$$

(7b)(iii)

The ball bearing is behaving like the bob of a pendulum with length l equal to the radius r of the semi-circular saucer.

Using

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{l}{g}}, \text{ we have}$$

$$T = 2\pi \sqrt{\frac{0.25}{9.81}}$$

$$T \approx 1.0 \text{ s}$$

7(c)(i)

An ideal gas obeys the Ideal Gas equation $pV = nRT$ where p = gas pressure, V = gas volume, n = number of moles of gas, R = molar gas constant, and T = thermodynamic temperature.

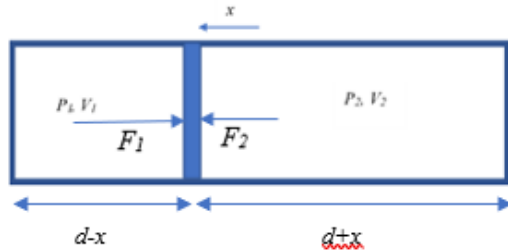
It is a gas in which inter-molecular forces of interaction are negligible.

7(c)(ii)

From $pV = nRT$, we have $pV = \text{constant}$ (Boyle's Law)

The gas must have a fixed mass and the temperature must remain constant for a change of volume and pressure.

7(c)(iii)



7(c)(iv)

Boyle's Law: $P_1V_1 = P_2V_2 = PV$

$P = \frac{F}{A}$ and $V = Ad$

$$\therefore F_1(d-x) = F_2(d+x) = PAd$$

7(c)(v)

By N2L, $F_2 - F_1 = ma$

$$\begin{aligned} F_1 \left(\frac{d-x}{d+x} \right) - F_1 &= ma \\ F_1 \left(\frac{d-x-d-x}{d+x} \right) &= ma \\ \frac{PAd}{(d-x)} \left(\frac{-2x}{d+x} \right) &= ma \end{aligned}$$

$$\therefore \frac{-2PAd}{d^2-x^2} x = ma$$

7(c)(vi)

When $d \gg x$, we have

$$a = \frac{-2PAd}{md^2} x$$

Since $a \propto -x$, the motion is simple harmonic, with $\omega^2 = \frac{2PA}{md}$

$$\text{The period is } T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{md}{2PA}}$$

7(c)(vii)

$$T = 2\pi \sqrt{\frac{(1.0)(0.50)}{2(100)(100 \times 10^{-4})}}$$

$$\approx 3.1 \text{ s}$$

$$0.8(-)(\cdot)$$